

Benefiting From OOD Samples in Open-Set Semi-Supervised Object Detection

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Abstract—Open-set semi-supervised object detection (OSSOD) is an emerging research area that relaxes the assumption of closed-set in semi-supervised object detection (SSOD), allowing unlabeled data to contain both in-distribution (ID) and out-of-distribution (OOD) samples. In the face of OSSOD, existing approaches expect to completely filter out OOD samples from unlabeled data for conventional semi-supervised learning. Different from it, we find that the effective use of OOD samples in unlabeled data can also facilitate feature learning and thus boost the detection performance of ID categories under open set conditions. Specifically, we perform refined instance-level consistency regularization (ICR) on all the detected instances (including the OOD ones) on unlabeled images, while introducing a novel OOD-aware contrastive learning (OCL) that clusters ID objects and pushes away OOD samples in the intra-class feature space, enhancing the compactness of ID features while emphasizing the discrimination between ID and OOD objects. Based on the distinguished features, we further design a prototype-based multimetric adaptive matching (MAM) approach, which identifies ID/OOD samples by adaptively measuring multiscale feature similarity between samples and class prototypes according to the specific category. With the help of this approach, more reliable ID/OOD samples can be mined from unlabeled data for contrastive learning, while the consistency regularization (CR) can be weighted according to the matching scores, correcting the biasing effects of OOD objects. Extensive experiments demonstrate that our design can effectively utilize OOD samples to optimize feature learning while avoiding their detrimental effects in semi-supervised learning, which ultimately greatly improves the detection capability of the model in open-set scenarios by a large margin over current state-of-the-art works.

Index Terms—Consistency regularization (CR), object detection, open-set problem, semi-supervised learning.

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I. INTRODUCTION

SEMI-SUPERVISED object detection (SSOD) focuses on boosting the effectiveness of object detectors by utilizing vast quantities of unlabeled data, which significantly mitigates the extensive time and labor required for instance-level annotation and is essential for contemporary deep neural models in detection tasks. While SSOD has achieved notable success [1], [2], [3] and attracted widespread attention, the prevalent assumption in this field is that unlabeled data share the same label space as labeled data. This closed-set assumption is problematic in real-world applications, as it involves extensive effort to verify whether the unlabeled data contains out-of-distribution (OOD) objects. To address this limitation, open-set semi-supervised object detection (OSSOD) has been introduced [4], offering a more realistic and challenging setup. OSSOD enhances detector optimization by utilizing unconstrained unlabeled images—including unseen OOD samples—alongside labeled data.

Addressing the open-set SSOD challenge is vitally important for practical applications, as it can considerably lessen the data preparation efforts. For instance, in autonomous driving [5], detection systems are typically trained on a fixed set of known categories, such as cars, pedestrians, and traffic lights. However, during real-world operation, vehicles inevitably encounter a variety of unlabeled objects, including both familiar and novel OOD instances—such as unusual road debris or animals not present in the training data. A similar challenge arises in automated retail checkout systems [6], where the model is trained on a predefined product catalog, yet frequently encounters new or seasonal items. In such cases, an effective OSSOD system can make use of these unlabeled samples to improve the detection of known categories without requiring exhaustive manual annotations, while also being robust to OOD inputs. This dual capability enhances both the safety and the adaptability of modern intelligent systems.

An intuitive approach to OSSOD, as seen in prior works, is to treat OOD samples as noise and to filter them out entirely before applying conventional SSOD pipelines. For instance, methods like [4] rely on a pretrained offline detector to eliminate OOD samples, which not only introduces significant computational overhead but also risks discarding potentially useful information. Critically, this filtering paradigm overlooks the fundamental opportunity that OOD samples present: they can be a valuable resource for learning more discriminative feature representations. Our work is motivated by a paradigm shift: instead of merely filtering OOD samples, we propose to

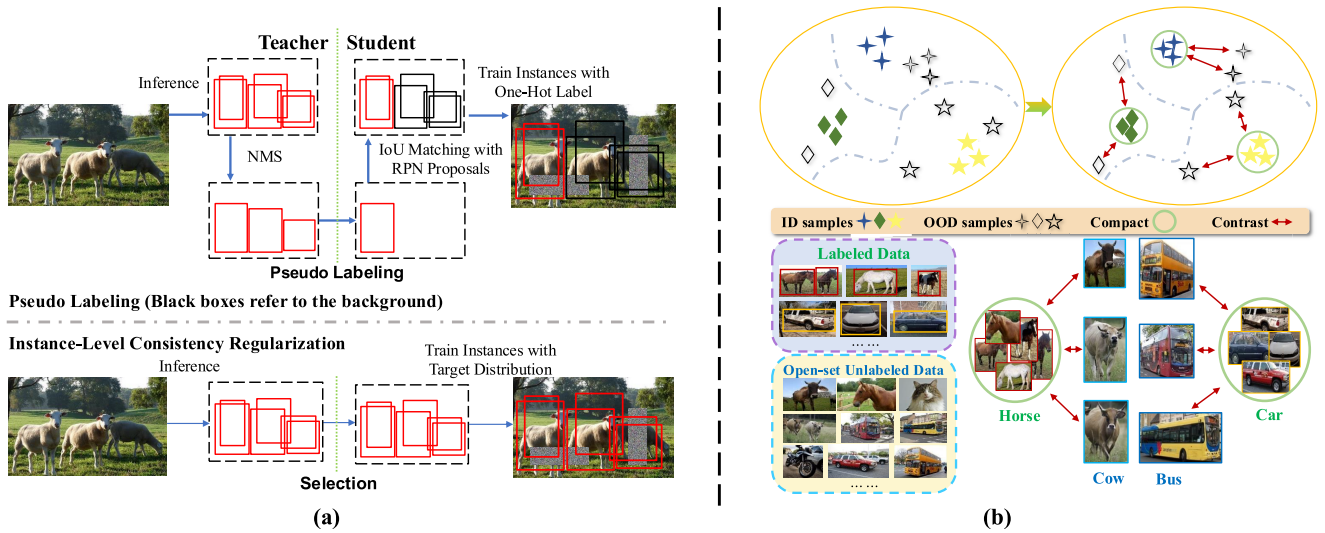


Fig. 1. Depiction of our proposed method. (a) Instance-level CR selects appropriate training instances directly from the Teacher model's inference results before NMS, allowing the model to overcome the limitations of pseudo-labels and enrich the foreground objects involved in training. (b) OCL clusters objects within the same ID category (e.g., horses) while separating OOD objects (e.g., cows in open-set unlabeled data), enhancing intra-class feature distinction and reducing the misclassification of OOD objects as known categories with high confidence.

effectively utilize them to improve the detector's robustness and ID object recognition capabilities. We argue that by explicitly modeling the distinction between ID and OOD objects in the feature space, the detector can learn to avoid overconfident misclassifications and sharpen its understanding of known categories.

To realize this vision, our work introduces a novel framework that integrates three key components. First, our proposed method, diverging from conventional pseudo-labeling in SSOD, employs instance-level consistency regularization (ICR) on unlabeled images. As illustrated in Fig. 1(a), this approach, effective in open-set contexts, enhances feature representation by learning invariance across weak and strong augmentations of objects, including OOD ones. It allows for the incorporation of potential ID objects with low confidence into the learning process, as opposed to their exclusion as background in pseudo-labeling frameworks. Empirical evidence supports the effectiveness of our method; applying instance-level consistency regularization (CR) enhances the model's detection capabilities and demonstrates the robustness of consistency learning. To further leverage OOD samples for developing more distinct feature representations at the instance level, we introduce an OOD-aware contrastive learning (OCL) head in the detector, which diverges from typical inter-class comparisons of supervised contrastive learning [7], [8], [9] by focusing on constructing the intrinsic structure of ID and OOD features within the intra-class space, as showcased in Fig. 1(b), thereby improving the efficiency of ID/OOD classification. Specifically, we derive contrastive loss within the intra-class feature space for each category, clustering ID objects while distancing corresponding OOD samples. In this context, "OOD" is relatively defined; for example, in a labeled image, a laptop is considered a "TV-like" OOD object relative to the TV category. For an unlabeled image, we identify the most relevant class for a detected object from the classification branch's predictive distribution and compare it to the class prototype to ascertain its ID or OOD status.

Implementing contrastive learning in the intra-class space, we introduce a novel class-aware, multimetric matching approach for adaptively distinguishing ID/OOD objects in each category within unlabeled images. Specifically, it maintains global prototypes for each ID category and compares the similarity between each instance and its corresponding category prototype from multiple perspectives. In addition, for OOD samples with low similarity, we reduce the weight of their CR to protect the integrity of learned ID category knowledge. Leveraging OCL coupled with our multimetric adaptive matching (MAM) mechanism enables the detector to effectively utilize both ID and OOD objects of each class, optimizing feature learning where OOD samples accentuate the distinction between ID and OOD features. By explicitly widening the gap between ID and OOD objects, our approach significantly reduces the likelihood of misclassifying unknown objects as known with high confidence, culminating in a marked enhancement of detector performance.

Our key contributions are as follows.

- 1) We introduce a novel framework that benefits from OOD samples in OSSOD, showing that appropriately utilizing OOD samples can enhance feature learning, leading to significant performance improvements in OSSOD.
- 2) We propose an instance-level CR approach and an OCL method that improves feature representation by distinguishing between ID and OOD features within each class.
- 3) We design a prototype-based, MAM mechanism that allows for the identification of ID and OOD samples in unlabeled data, further optimizing the feature learning process and improving model robustness.
- 4) Extensive experimental results on benchmark datasets demonstrate the effectiveness of the proposed method, with its performance significantly surpassing existing state-of-the-art OSSOD algorithms.

Noticeably, the recent work in the OSSOD field, CFL-Detector [10], also utilizes contrastive learning. However,

unlike CFL-Detector, our method incorporates a prototype-based MAM mechanism using EMA, while CFL-Detector focuses on pure feature clustering and logits uncertainty learning. In Section IV-B, we include comparative experiments, which demonstrate that our approach provides a significant advantage in OOD detection.

II. RELATED WORK

OSSOD, as proposed by Liu et al. [4], aims to enhance the capability to detect in-distribution (ID) objects with unlabeled data that comprises OOD samples. As opposed to open-vocabulary [11], [12], [13], [14], [15] or open-world [16], [17], [18], [19] object detection, OSSOD seeks to get rid of potentially negative impacts from OOD samples.

A. Semi-Supervised Object Detection

With the advent of effective methods [20], [21], [22] in semi-supervised image classification, SSOD predominantly leverages two strategies: pseudo-label generation and CR. Pseudo-labels-based methods estimate the pseudo-labels for unlabeled images by a pretrained model and jointly train the model with augmented labeled and unlabeled data, which can be refined as self-training [23], [24], [25], [26], [27], mean-teacher [28], [29], soft labeling [30], [31], [32], dense guidance-based [33], [34], uncertainty quantification [35], [36], and other optimized pseudo-labels methods [2], [37], [38], [39]. Among these, the recent work named USD [39] leverages an uncertainty-based one-stage learning framework to effectively mitigate false-negative (FN) and false-positive (FP) issues during pseudo-label generation, enhancing pseudo-label reliability. Additionally, CR based methods [40], [41] aim to regularize the consistency of the prediction for the same unlabeled images under various data augmentations. Nonetheless, each of the aforementioned methods is predicated on a closed-set assumption. In contrast, our approach contemplates a more realistic scenario wherein the unlabeled data encompasses novel objects that have not been observed previously.

B. Open-Set Semi-Supervised Learning

The primary focus of current open-set semi-supervised learning algorithms is the image classification task [42], [43], [44], [45], [46], [47], [48], [49], [50]. To lessen the negative effects of OOD, Yu et al. [48] iteratively assign trustworthy pseudo-labels to the unlabeled data; Open-match [46] maintains a binary classifier for each class to identify OOD samples. Huang et al. [47] employ OOD samples for self-supervised learning and concurrently filter them out to boost feature extraction efficiency; and Luo et al. [42] discover that consistency learning on OOD samples can also help the model. Zhao et al. [50] address open-world semi-supervised learning by introducing a unified framework that leverages the Sinkhorn–Knopp algorithm for balanced pseudo labeling and employs a mixing policy to facilitate knowledge transfer across seen and novel classes. The aforementioned techniques are intended for classification tasks, with the OSSOD problem remaining an active research area. Liu et al. [4] introduced an

offline detector for filtering OOD pseudo-labels in OSSOD, which, however, incurs high computational costs and shows limited performance. Our approach, in contrast, incorporates an OCL head into the primary detector, enhancing the model’s capability to manage OOD samples and yielding substantial improvements.

C. Open-Set Object Detection

Open-set object detection aims for models to recognize open-set objects during testing. The “overlooked elephant” phenomenon [51] illustrates how object detectors, when trained on labeled data, often misclassify OOD samples as high-confidence ID objects, limiting their robustness. Various methods have been proposed to counter this, including one-vs-rest classifiers for OOD identification [52] and uncertainty-guided dropout for OOD filtering [53]. Based on open-set object detection, open-vocabulary object detection [11], [12], [13], [14], [15], [54] requires the simultaneous identification of both known and unknown classes during testing, with the ability to distinguish between unknown classes. For instance, a recent approach, VTP-OVD [15], introduced a fine-grained visual-text prompt-driven self-training paradigm. By enhancing dense alignment between visual and textual prompts, this method significantly improves the reliability of pseudo-labels in open-vocabulary object detection, thus effectively addressing the challenge of distinguishing known from unknown classes. More recently, open-world object detection [17], [18], [55], [56], [57] have emerged as a more demanding task, integrating novel class discovery with open-set recognition. Open-world object detection involves classifying OOD samples, with notable methods like ORE [18] using contrastive loss for enhanced OOD differentiation, and UC-OWOD [17] employing a “ $K + U$ ”-way classifier for detection and clustering. Additionally, another recent approach, MEPU [57], tackles the label bias problem in open-world object detection (OWOD) by proposing an unsupervised strategy for modeling and extending pseudo-unknown objects. This approach significantly improves the detection performance for unknown classes while maintaining the detection accuracy for known categories, further advancing research in OWOD tasks. In this work, OSSOD, however, focuses on refining the model’s discrimination of known objects using unlabeled images containing open-set samples.

III. METHODOLOGY

A. Overview

OSSOD aims to train an object detector with a small labeled dataset $\mathcal{D}_l = \{(\mathbf{x}_i^l, \mathbf{y}_i^l)\}_{i=1}^{N_l}$ of size N_l alongside a large unlabeled dataset $\mathcal{D}_u = \{(\mathbf{x}_i^u)\}_{i=1}^{N_u}$ of size N_u ($N_u \gg N_l$). The category labels in the annotation $\mathbf{y}_i^l$ of $\mathbf{x}_i^l$ belong to $\mathcal{C}^l = \{1, \dots, C\}$, while the category labels of $\mathbf{x}_i^u$ belong to its superset $\mathcal{C}^u$ ($\mathcal{C}^l \subset \mathcal{C}^u$), meaning that objects from novel categories, i.e., OOD objects, are involved in $\mathcal{D}_u$. As such, existing SSOD methods based on the closed-set assumption do not generalize well to OSSOD, where addressing OOD objects in unlabeled datasets is crucial. Instead of using an intuitive solution that completely eliminates the OOD samples [4], we propose in this work to utilize

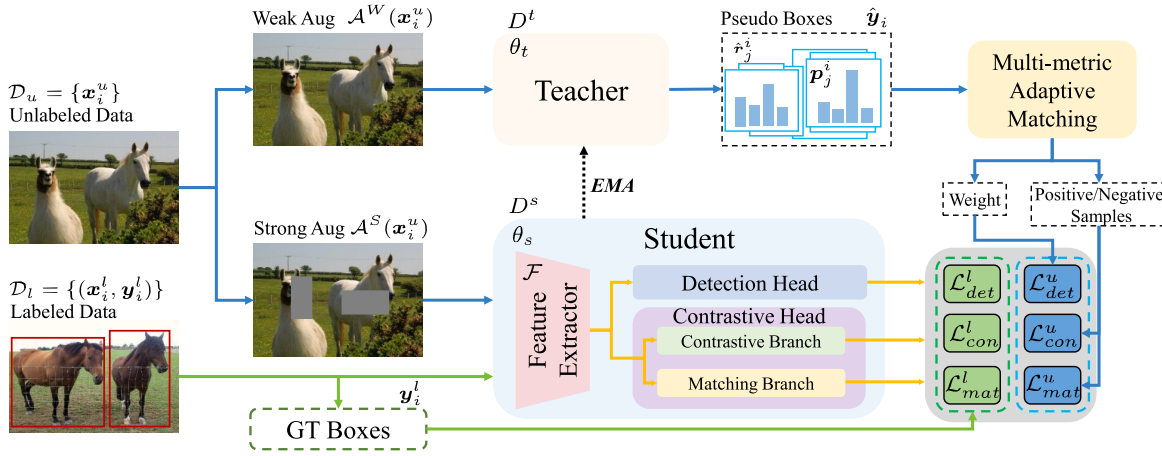


Fig. 2. Overview of the proposed method. Enhancing the conventional Teacher–Student semi-supervised paradigm, our approach introduces a contrastive head in parallel to the detection head, specifically targeting OOD samples in unlabeled images. This is achieved through three key components: a refined instance-level CR for feature invariance learning, an OCL strategy to explicitly distinguish between ID and OOD objects, and an MAM algorithm to accurately identify ID/OOD objects in unlabeled data.

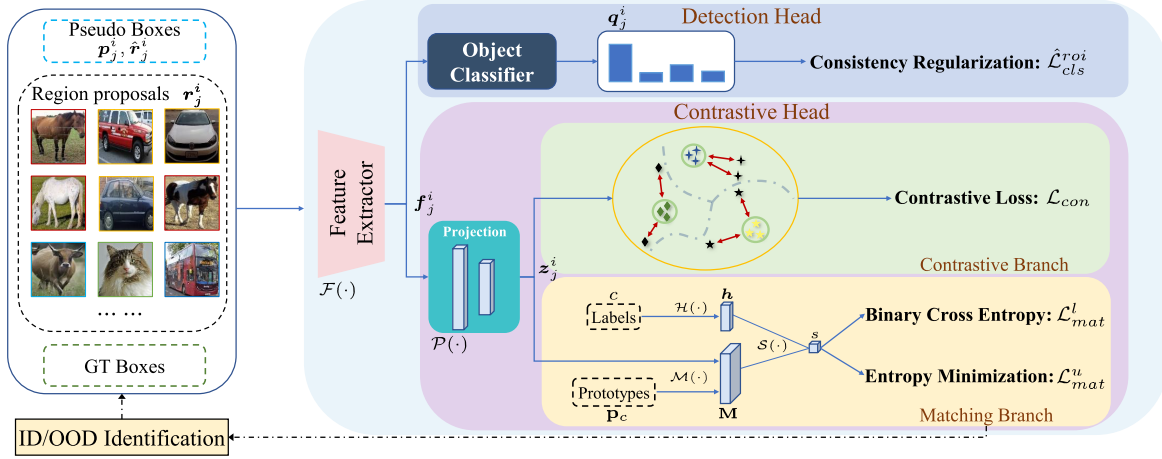


Fig. 3. Detailed pipeline of our proposed method. For the classification branch in the detection head, instance-level CR is enforced to strengthen the feature extraction ability of the model. The contrastive learning branch is aimed at further utilizing the OOD samples to optimize the intra-class feature space, where the discrimination between the ID and OOD objects is emphasized. The matching branch uses the class-specific information and correlation with prototypes to identify ID/OOD samples in unlabeled data.

them for better feature learning, so as to enhance the model's ability to detect both ID and OOD objects. Fig. 2 provides an overview of the proposed method, while Fig. 3 illustrates its detailed pipeline.

To this end, we apply a self-training paradigm to the OSSOD problem as it achieves notable performance on SSOD tasks as in [58]. Specifically, a Teacher–Student framework [59] maintaining two independent Faster R-CNN [60] detectors D^t and D^s is used as our base structure, with the Student model's parameter θ_s being updated by optimizing the objective function, and the Teacher model's parameter θ_t being accumulated by exponential moving average (EMA) [59] by $\theta_t = \varepsilon\theta_t + (1 - \varepsilon)\theta_s$. After initialized training of the framework with supervised learning on labeled images, we then incorporate the unlabeled images for semi-supervised learning.

At the semi-supervised learning stage, the loss of normal supervised training on the labeled dataset $\mathcal{D}_l = \{(\mathbf{x}_i^l, \mathbf{y}_i^l)\}$ is defined as follows:

$$\mathcal{L}_{det}^l(\mathbf{x}_i^l, \mathbf{y}_i^l) = \sum_{\mathbf{L} \in \mathbf{L}} \mathcal{L}(\mathbf{x}_i^l, \mathbf{y}_i^l) \quad (1)$$

where $\mathbf{L}$ is a set of four typical losses on classification and regression involving region-proposal network (RPN) and region-of-interest (ROI) head in Faster-RCNN, i.e., $\mathbf{L} = \{\mathcal{L}_{cls}^{rpn}, \mathcal{L}_{reg}^{rpn}, \mathcal{L}_{cls}^{roi}, \mathcal{L}_{reg}^{roi}\}$. For the unlabeled dataset $\mathcal{D}_u = \{\mathbf{x}_i^u\}$, pseudo-labels $\hat{\mathbf{y}}_i$ are generated by the Teacher model D^t with the weakly augmented versions $\hat{\mathbf{x}}_i^u$ to train the Student model D^s fed with the strongly augmented versions $\tilde{\mathbf{x}}_i^u$ in a supervised manner similarly, except that the loss function set $\mathbf{L}$ is modified via replacing $\mathcal{L}_{cls}^{roi}$ by a new loss $\hat{\mathcal{L}}_{cls}^{roi}$. Mathematically, the loss on $\mathcal{D}_u = \{\mathbf{x}_i^u\}$ can be written as follows:

$$\begin{aligned} \mathcal{L}_{det}^u(\mathbf{x}_i^u) &= \sum_{\mathbf{L} \in \hat{\mathbf{L}}} \mathcal{L}(\tilde{\mathbf{x}}_i^u, \hat{\mathbf{y}}_i) \\ \hat{\mathbf{y}}_i &= D^t(\hat{\mathbf{x}}_i^u), \quad \hat{\mathbf{x}}_i^u = \mathcal{A}^W(\mathbf{x}_i^u), \quad \tilde{\mathbf{x}}_i^u = \mathcal{A}^S(\mathbf{x}_i^u) \end{aligned} \quad (2)$$

where $\hat{\mathbf{L}} = \{\mathcal{L}_{cls}^{rpn}, \mathcal{L}_{reg}^{rpn}, \hat{\mathcal{L}}_{cls}^{roi}, \mathcal{L}_{reg}^{roi}\}$, and $\mathcal{A}^W, \mathcal{A}^S$ represent weak and strong augmentation, respectively. The new loss $\hat{\mathcal{L}}_{cls}^{roi}$ is defined for robust instance-level CR to strengthen the capability of feature extraction, as detailed in Section III-B. $\mathcal{L}_{det}^l$ and $\mathcal{L}_{det}^u$ compose the overall semi-supervised detection

loss $\mathcal{L}_{det}$ by a balancing parameter λ , i.e., $\mathcal{L}_{det} = \mathcal{L}_{det}^l + \lambda \mathcal{L}_{det}^u$. Furthermore, to better utilize the OOD samples to optimize feature learning, we introduce a contrastive head parallel to the detection head in each Faster-RCNN model D^l and D^s , including a contrastive branch and a matching branch. The former one aims to clusters the ID objects and pushes OOD samples away at the intra-class feature space via a contrastive loss $\mathcal{L}_{con}$ involved (Section III-C), and the latter one undertakes the task of identification of ID/OOD for each category via a matching loss $\mathcal{L}_{mat}$ to achieve the contrastive learning (Section III-D). As a result, the entire loss for our proposed method can be formulated as

$$\mathcal{L} = \mathcal{L}_{det} + \gamma \mathcal{L}_{con} + \mathcal{L}_{mat} \quad (3)$$

where γ is the balancing parameter that controls the tradeoff between the loss terms.

B. Instance-Level CR

Existing SSOD studies widely adopt the pseudo-labeling mechanism [58] that has achieved great success in semi-supervised image classification tasks based on the closed-set setting, which is incompatible with OSSOD for two reasons. One is that the detector can easily misidentify OOD samples as ID classes with high confidence, causing an accurate OOD detector to be unavoidable for pseudo-label postprocessing, such as the existing OSSOD method [4], which adopts an offline self-training DINO [61], so as to cost extra computational resources significantly. Another one is that, despite the computational resources, the involved OOD detector may be of unreliable recall of pseudo labels, causing ID objects to be misidentified as background to corrupt the training process.

Based on the above considerations, we propose to apply refined instance-level CR to OSSOD instead of the conventional pseudo-labeling scheme, as shown in Fig. 1. Unlike conventional SSOD methods, which discard OOD samples, our approach retains these samples, preventing the risk of missing valuable information. To help clarify our approach, consider an example: in satellite imagery for crop monitoring, new crop varieties may appear during certain seasons. These unlabeled crops are typically excluded from closed-set SSOD frameworks. However, our method allows these potential OOD crops to be incorporated into the training process, enabling the model to learn more distinguishable feature representations and improve detection accuracy for both known and novel crop types.

Technically, the Teacher model D^l produces a series of region proposals $\hat{r}_j^i$ with its RPN as “instances” given an image $\hat{x}_i^u$, i.e., the weakly augmented version of an unlabeled image x_i^u , each of which associates a target distribution p_j^i obtained by the ROI head of D^l . Note that these proposals $\hat{r}_j^i$ are densely selected before NMS, including all foreground boxes and randomly selected background boxes for sufficient sampling. Meanwhile, for each proposal $\hat{r}_j^i$, the Student model D^s predicts a distribution q_j^i on the strongly augmented version $\tilde{x}_i^u$. With the two distributions computed, we train D^s using p_j^i as the supervision with a typical soft cross-entropy loss. So

TABLE I

COMPARISON OF THE PROPOSED METHOD AND THE CSD METHOD [40] ON CONSISTENCY LEARNING ACROSS TWO MECHANISMS

Method	Labeled	Unlabeled	mAP
Supervised baseline in [40] CSD [40]	VOC07	-	73.9
	VOC07	VOC12	74.7 (+0.8)
Faster R-CNN Ours	VOC07	-	76.3
	VOC07	VOC12	81.5 (+5.2)

for an unlabeled image x_i^u , an instance-level consistency loss is defined as follows:

$$\begin{aligned} \hat{\mathcal{L}}_{cls}^{roi}(\hat{x}_i^u, \hat{y}_i) &= \frac{1}{N_r} \sum_{j=1}^{N_r} w_j \cdot \ell(\hat{r}_j^i) = \frac{1}{N_r} \sum_{j=1}^{N_r} w_j \cdot \ell(p_j^i, q_j^i) \\ \ell(p_j^i, q_j^i) &= \sum_{c=1}^C -p_{j,c}^i \log q_{j,c}^i \end{aligned} \quad (4)$$

where $\hat{y}_i = D^l(\hat{x}_i^u) = \{\hat{r}_j^i, p_j^i\}$, $q_j^i = D^s(\hat{x}_i^u, \hat{r}_j^i)$, and $N_r = 512$ is the default number of training boxes in Faster R-CNN. With this loss introduced, instance-level CR takes effect as a soft constraint against pseudo-label as a hard constraint for all region proposals, which guides the model to learn the feature invariance of the objects under different views without abandoning potential OOD samples. It properly utilizes the relatively low-confidence objects instead of misidentifying them as background. However, the overlearning of OOD samples could destroy the learned knowledge of the known classes of ID samples, so we employ matching similarity scores w_j to weight the consistency loss of the OOD samples, which greatly reduces the detrimental effects of the OOD samples and stabilize the training, as detailed in Section III-D.

It is worth noting that CSD [40] also performs similar consistency learning as us, while it works on the flipped detection boxes on flipped images rather than establishing correspondence of detection boxes across different augmented images and models like us. One further significant distinction is that the proposed method has greater flexibility in choosing training boxes. CSD filters training boxes around high-confidence pseudo-labels and lacks the utilization of potential foreground features in images. Our method can achieve much greater performance by mining and utilizing these potential and dependable foreground features. It is observed in Table I that even with a baseline model (Faster R-CNN w/ Resnet50-FPN) that has higher performance, the proposed refined CR method still achieves a significant improvement in SSOD, improving performance from a supervised baseline by + 5.2 mAP compared with + 0.8 mAP of CSD.

C. OOD-Aware Contrastive Learning

We further propose a contrastive learning method to deal with the presence of OOD objects in the OSSOD task, considering the unawareness of OOD samples by instance-level CR. The core idea of this method is to design an OCL mechanism, which focuses on clustering known objects while pushing OOD objects apart in the feature space, thereby improving the distinction between ID and OOD samples. This approach is similar to applications in large-scale security

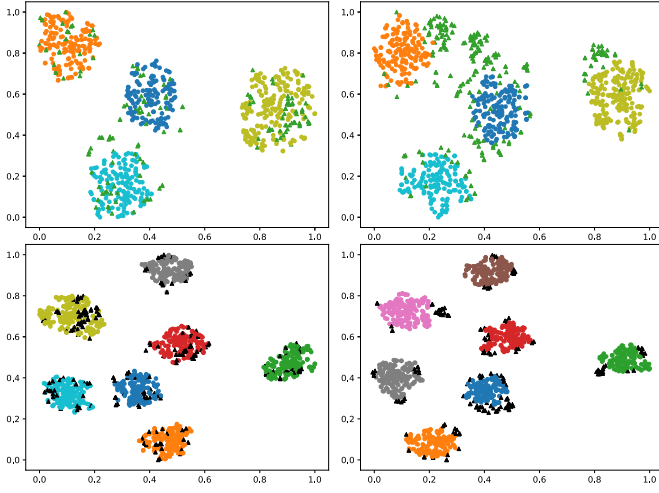


Fig. 4. Visualization of the learned features without (left) and with (right) the proposed OOD-aware contrastive loss. ID features are represented by circles, while OOD features are represented by triangles.

systems, such as airport surveillance. In such cases, the model needs to distinguish new, previously unseen objects (e.g., unusual luggage or new types of vehicles) from familiar categories, thus preventing these novel objects from being misclassified as background or known classes, significantly enhancing detection performance. To achieve this, an OCL head is designed within the ROI head of Faster R-CNN in par with the detection head (classification and regression), to distinguish the ID/OOD feature distribution at the intra-class space, as illustrated in Fig. 3.

For a foreground region proposal r_j^i from the Student model D^s given x_i ($x_i \in \{x_i^u, x_i^l\}$), the feature extractor $\mathcal{F}(\cdot)$ of Faster R-CNN produces an ROI feature f_j^i that is further projected to a contrastive feature $z_j^i \in \mathbb{R}^{D_c}$ by a multilayer perceptron (MLP) $\mathcal{P}(\cdot)$ composed of sequential FC, ReLU, FC layers, i.e., $z_j^i = \mathcal{P}(f_j^i) = \mathcal{P}(\mathcal{F}(r_j^i))$. Meanwhile, with the classification distribution q_j^i of r_j^i produced by the original detection head, we can easily obtain its predicted category $\hat{c}_j^i$ by $\hat{c}_j^i = \arg \max_c q_{j,c}^i$, with which we identify r_j^i as an ID or OOD sample following two cases. For labeled images x_i^l , if $\hat{c}_j^i$ fails to match the ground truth class c_j^i recorded in y_i^l , including mis-classified category or no object, it is referred to an OOD sample associated with the predicted category $\hat{c}_j^i$, i.e., $r_j^i \in \mathbb{C}_{\hat{c}_j^i}^{ood}$, otherwise it is regarded as an ID object, i.e., $r_j^i \in \mathbb{C}_{c_j^i}^{id}$. For unlabeled images x_i^u , we design a MAM approach for identification, as detailed in Section III-D. In either case, a region r_j^i is identified as an ID or OOD sample with respect to a specific class $\hat{c}_j^i$, so that a contrastive loss is defined as

$$\mathcal{L}_{con}(x_i) = \frac{1}{N_{id}} \sum_{j=1}^{N_{id}} \varphi(r_j^i) = \frac{1}{N_{id}} \sum_{j=1}^{N_{id}} \varphi(z_j^i)$$

$$\varphi(z_j^i) = \frac{-1}{|\mathbb{C}_{\hat{c}_j^i}^{id}|} \sum_{k=1, k \neq j}^{|\mathbb{C}_{\hat{c}_j^i}^{id}|} \log \frac{\exp(\hat{z}_j^i \cdot \hat{z}_k^i / \tau)}{\sum_{v=1, v \neq j}^{|\mathbb{C}_{\hat{c}_j^i}^{id}|} \exp(\hat{z}_j^i \cdot \hat{z}_v^i / \tau)} \quad (5)$$

where τ is the temperature for contrastive learning, N_{id} is the number of ID objects, and $\hat{z}_j^i$ is the L2-normalized z_j^i , hence $\hat{z}_j^i \cdot \hat{z}_k^i$ measures the cosine similarity between instances. $\mathbb{C}_{\hat{c}_j^i}^{id}$

$\mathbb{C}_{\hat{c}_j^i}^{id} \cup \mathbb{C}_{\hat{c}_j^i}^{ood}$ is the set of instances that are assigned to the same class $\hat{c}_j^i$ as r_j^i . In practice, we calculate the loss within a batch of images for the richness of ID and OOD samples. With the loss in (5) defined, the proposed approach can explicitly cluster ID objects of the same class together while enlarging the distance between ID and OOD objects, as illustrated in Fig. 4. As a result, it improves the detection performance by + 1.75 gain in mAP, as shown in Table II (Row 3 compared with Row 1).

D. Multimetric Adaptive Matching

In order to mine reliable ID/OOD samples from unlabeled images for intra-class contrastive learning, a prototype-based MAM branch is devised by leveraging information from class prototypes and class label vectors.

To better illustrate this approach, we can refer to an example from robotics. In robotics, suppose a robot is trained to pick specific tools. However, it may encounter new types of tools that it has never seen before. By using MAM, the robot can identify these new tools as OOD objects and adjust its behavior accordingly. At the same time, it can continue to perform tasks using familiar tools, thereby enhancing its ability to adapt to a wide range of situations in dynamic environments.

Specifically, for each known class $c \in \mathcal{C}^l$, we maintain a global prototype $\mathbf{p}_c$ for it as in [3], being initialized with Gaussian noise and updated with the optimized contrastive features z_c of ground truth proposals in labeled data via EMA algorithm.

The EMA updating strategy progressively stabilizes prototype evolution during training by smoothly averaging prototype features across multiple iterations, effectively alleviating the adverse impact of class imbalance in labeled data on prototype quality, and thereby ensuring more reliable mining of ID/OOD samples. Then, we define a matching score $s(\mathbf{r}, c)$ ranging from (0, 1) to measure how close it is from $\mathbf{r}$ to a specified class c , which is achieved by several fully connected layers. To be specific, given a pair of $\mathbf{r}$ and c , $s(\mathbf{r}, c)$ is computed as follows:

$$s(\mathbf{r}, c) = \mathcal{S}(\mathbf{M} \cdot \mathbf{h}) = \mathcal{S}(\mathcal{M}(\mathbf{z}, \mathbf{p}_c) \cdot \mathcal{H}(c)) \quad (6)$$

where $\mathbf{M} \in \mathbb{R}^{D_c \times D_M}$ is a metric matrix produced by an FC layer $\mathcal{M}$ fed with the concatenation of $\mathbf{z}$ and $\mathbf{p}_c$, $\mathbf{h} \in \mathbb{R}^{D_M}$ encodes the class-specific attention transformed from the one-hot vector of c via a block of 2 FC layers $\mathcal{H}$, and $\mathcal{S}$ is a linear layer followed by a sigmoid function. To train the matching branch as above, we define a matching loss $\mathcal{L}_{mat}$ following two cases for a given image x_i^l or x_i^u . For x_i^l , as the ground-truth class c_j^i of a region instance r_j^i is available, we compute three matching scores $s(r_j^i, c)$, $c \in \{c_j^i, \bar{c}_j^i, c_j^{*i}\}$, where $\bar{c}_j^i$ refers to a randomly selected class label other than c_j^i , and c_j^{*i} is the class label with the largest predicted probability within all $\hat{c}_j^i$, i.e., the hardest class. Then, we utilize the binary cross-entropy as the matching loss, i.e.,

$$\mathcal{L}_{mat}^l(x_i^l) = \frac{1}{N_r} \sum_{j=1}^{N_r} \psi(r_j^i)$$

$$\psi(r_j^i) = -(\zeta(r_j^i, c_j^i) + \eta(r_j^i, \bar{c}_j^i) + \eta(r_j^i, c_j^{*i})) \quad (7)$$

TABLE II

ABLATION STUDY RESULTS ON THE KEY COMPONENTS OF THE PROPOSED METHOD: **CR**, CONTRASTIVE LEARNING ON LABELED DATA ONLY (**CON-L**) OR ON BOTH LABELED AND UNLABELED DATA (**CON-FULL**), AND **MAM**. EACH VARIATION OF THE PROPOSED METHOD IS BUILT UPON THE “**BASELINE**” MODEL, A PREVIOUS CLASSIC SSOD METHOD [28]

#	Method	mAP
1)	baseline	20.98
2)	baseline + CR w/o w_j	22.94
3)	baseline + CON-L	22.73
4)	baseline + CR + MAM	24.41
5)	baseline + CON-full + MAM	24.68
6)	baseline + CR + CON-full + MAM w/o $\mathcal{L}_{mat}^u$	25.42
7)	baseline + CR + CON-full + MAM (Ours)	26.09

where $\zeta(\mathbf{r}, c) = \log(s(\mathbf{r}, c))$ and $\eta(\mathbf{r}, c) = \log(1 - s(\mathbf{r}, c))$. For $\mathbf{x}_i^u$ which has no ground-truth labels, we compute two matching scores $s(\mathbf{r}_j^i, c), c \in \{\hat{c}_j^i, \tilde{c}_j^i\}$, where $\tilde{c}_j^i$ refers to a randomly selected class label other than the predicted pseudo label $\hat{c}_j^i$. We further define a matching loss based on entropy minimization [62], considering the unreliability of pseudo label in unlabeled data, i.e.,

$$\mathcal{L}_{mat}^u(\mathbf{x}_i^u) = \frac{1}{N_r} \sum_{j=1}^{N_r} \xi(\mathbf{r}_j^i) \quad (8)$$

$$\xi(\mathbf{r}_j^i) = -(\varrho(\mathbf{r}_j^i, \hat{c}_j^i) + \varrho(\mathbf{r}_j^i, \tilde{c}_j^i)).$$

Here, $\varrho(\mathbf{r}, c) = s(\mathbf{r}, c) \log(s(\mathbf{r}, c)) + (1 - s(\mathbf{r}, c)) \log(1 - s(\mathbf{r}, c))$. As such, the matching branch provides a comprehensive measure of an object’s proximity to the prototype through multimetric and class-specific adaptive information. Objects with matching scores above σ_l are marked as class-wise ID objects, and ones below σ_s are marked as class-wise OOD objects for contrastive learning, while the ambiguous features are ignored due to their uncertainty. Here, σ_l and σ_s are predefined thresholds to filter out samples. Besides, we also weight the CR loss in (4) for foreground samples with low matching scores to reduce the accumulation of OOD knowledge that may confound known category identification, i.e., the weight w_j in (4) is denoted as follows:

$$w_j = \begin{cases} s(\mathbf{r}_j^i, \hat{c}_j^i), & \text{if } s(\mathbf{r}_j^i, \hat{c}_j^i) < \sigma_s \\ 1, & \text{otherwise} \end{cases} \quad (9)$$

where $\hat{\mathbf{r}}_j^i$ is the j th region proposal produced by Teacher model D^t given unlabeled image $\hat{\mathbf{x}}_i^u$, and $\hat{c}_j^i$ is the pseudo-category label predicted by Student model D^s given $\hat{\mathbf{r}}_j^i$. With MAM, the features in unlabeled data are fully exploited by CR and contrastive learning, thus enhancing the model’s ability to detect both ID and OOD objects.

With all the losses defined, (3) is refined as follows:

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{det} + \gamma \mathcal{L}_{con} + \mathcal{L}_{mat} \\ &= \mathcal{L}_{det}^l + \lambda \mathcal{L}_{det}^u + \gamma (\mathcal{L}_{con}^l + \mathcal{L}_{con}^u) + \mathcal{L}_{mat}^l + \mathcal{L}_{mat}^u. \end{aligned} \quad (10)$$

Here, $\mathcal{L}_{con}^l$ and $\mathcal{L}_{con}^u$ are the contrastive loss calculated in labeled dataset $\mathcal{D}_l$ and unlabeled dataset $\mathcal{D}_u$, respectively. All of the losses are normalized by the respective number of images in the training data batch.

TABLE III

SELECTED ID CATEGORIES FOR DIFFERENT CATEGORY COUNTS

ID Category Count	Selected Categories
20	Person, Bicycle, Airplane, Truck, Cat, Horse, Cow, Giraffe, Baseball glove, Wine glass, Broccoli, Spoon, Knife, Pizza, Donut, Cake, TV, Mouse, Microwave, Refrigerator
40	Person, Bicycle, Airplane, Bus, Traffic light, Bench, Bird, Cat, Horse, Cow, Knife, Giraffe, Snowboard, Umbrella, Tie, Backpack, Baseball bat, Skateboard, Bottle, Wine glass, Fork, Bear, Bowl, Sandwich, Orange, Dount, Chair, Potted plant, Bed, Dining table, Toilet, Laptop, Remote, Cell phone, Microwave, Oven, Toaster, Book, Refrigerator, Clock
60	Person, Bicycle, Car, Motorcycle, Airplane, Bus, Train, Traffic light, Fire hydrant, Parking meter, Bench, Bird, Dog, Horse, Sheep, Cow, Bear, Zebra, Giraffe, Backpack, Umbrella, Handbag, Tie, Frisbee, Snowboard, Sports ball, Kite, Baseball bat, Baseball glove, Skateboard, Bottle, Cup, Wine glass, Fork, Knife, Spoon, Bowl, Sandwich, Orange, Broccoli, Hot dog, Dount, Chair, Bed, Potted plant, Dining table, Toilet, TV, Mouse, Mouse, Remote, Cell phone, Microwave, Oven, Toaster, Sink, Refrigerator, Book, Clock, Toothbrush

IV. EXPERIMENTS

A. Experimental Setups

Datasets: Similar to [4], we validate the superiority of the proposed method using two specific protocols: **COCO-Open** and **COCO-OpenImages**.

- 1) **COCO-Open:** This protocol is constructed based on the COCO2017-train dataset [69]. Specifically, 20, 40, or 60 classes are randomly selected as the ID classes, while the remaining classes are considered OOD. Additionally, we create a set of labeled data containing 1K (i.e., 1×10^3), 2K, and 4K images to evaluate the performance of our method under varying amounts of labeled data. The labeled samples are randomly drawn from the pure ID class set, excluding any OOD samples.
- 2) **COCO-OpenImages:** This protocol is designed for an in-depth study of OSSOD. In this protocol, the entire COCO2017-train [69] dataset (117K images) is used as labeled data, while the OpenImagesv5 dataset (1.7 M images) [70] serves as unlabeled data. All 80 classes in the COCO dataset are treated as known ID classes, while the OpenImagesv5 dataset contains 601 distinct object categories, most of which are OOD samples relative to the COCO dataset.
- 3) **VOC-COCO:** This protocol is designed to evaluate our proposed method in a more challenging cross-dataset OSSOD scenario. In this work, the labeled dataset is drawn from the Pascal VOC training set, containing 11K images from 20 object categories. Meanwhile, the unlabeled dataset is derived from the MS-COCO training set, which includes an additional 60 classes absent from VOC, treated as OOD categories during training.

Data Split: In constructing the **COCO-Open** protocol, the data must be split into ID and OOD categories. Because the categories are selected randomly, the splits in our experiments cannot exactly match those in [4]. To avoid unfair performance comparisons resulting from inconsistent ID and OOD splits, we conducted multiple random category selections and determined the final splits based on baseline performance. The

TABLE IV
SUMMARY OF DATA AUGMENTATION STRATEGIES USED IN OUR PROPOSED METHOD FOR LABELED AND UNLABELED IMAGES

Augmentation	Labeled Image Training	Unlabeled Image Training	weak	strong
Scale jitter	short edge $\in(0.5, 1.5)$	short edge $\in(0.5, 1.5)$	✓	✓
Solarize jitter	p=0.25, ratio $\in(0, 1)$	p=0.25, ratio $\in(0, 1)$	-	✓
Brightness jitter	p=0.25, ratio $\in(0, 1)$	p=0.25, ratio $\in(0, 1)$	-	✓
Contrast jitter	p=0.25, ratio $\in(0, 1)$	p=0.25, ratio $\in(0, 1)$	-	✓
Sharpness jitter	p=0.25, ratio $\in(0, 1)$	p=0.25, ratio $\in(0, 1)$	-	✓
Translation	-	p=0.3, translation ratio $\in(0, 0.1)$	-	✓
Rotate	-	p=0.3, angle $\in(0, 30^\circ)$	-	✓
Shift	-	p=0.3, angle $\in(0, 30^\circ)$	-	✓
Cutout	num $\in(1, 5)$, ratio $\in(0.05, 0.2)$	num $\in(1, 5)$, ratio $\in(0.05, 0.2)$	-	✓

average performance of baseline methods on these splits is similar to the results reported in [4]. To ensure the fairness and robustness of the evaluation, we use these splits in subsequent experiments. Table III presents the final splits and their corresponding category names for different numbers of ID categories.

Implementation Details: In this work, our proposed algorithm is implemented using the MMDetection framework [71],¹ an open-source object detection toolbox built on PyTorch.² Specifically, we adopt Faster R-CNN with a ResNet50-FPN backbone as the base detectors.

To ensure that *instance-level CR* effectively enables the model to learn feature invariances, we adopt the standard and well-established weak-strong augmentation strategy outlined in [28]. This strategy has been widely applied in tasks such as SSOD [29], [30], [37], where its effectiveness has been well-demonstrated. The specific implementation details of the augmentation strategy are provided in Table IV.

Following common practice in semi-supervised learning [58], [59], we utilize an EMA strategy with an EMA parameter $\varepsilon = 0.999$ to update both model parameters and class prototypes. The unsupervised detection loss $\mathcal{L}_{det}^u$ is weighted by $\lambda = 4.0$, aligning with commonly adopted configurations [28], [37]. Meanwhile, the weight coefficient of the contrastive loss $\mathcal{L}_{con}$ is set to $\gamma = 0.1$, determined by a grid search on a validation split to ensure each loss component contributes reasonably and proportionately during training. For contrastive learning and MAM, we set the temperature coefficient $\tau = 0.2$, the dimension of object representations to $D_C = 128$, thresholds to $\sigma_l = 0.75$ and $\sigma_s = 0.2$, and the dimension of the metric matrix to $D_M = 5$. These hyperparameters are further analyzed in the ablation study. Regarding the training configurations, for the **COCO-Open** and **VOC-COCO** datasets, the total number of training iterations is set to 180K, including a 40K iteration burn-in stage, with batch sizes of labeled and unlabeled data being 4 and 16, respectively. For the larger-scale **COCO-OpenImage** dataset, training iterations are increased to 720K (including 80K burn-in iterations), with batch sizes for both labeled and unlabeled data set to 32. Additionally, we use an SGD optimizer with a momentum of 0.9 and an initial learning rate of 0.01. During training, the learning rate is reduced by

a factor of 10 at iterations 120 and 160K for **COCO-Open** and **VOC-COCO**, and at iterations 480 and 680K for **COCO-OpenImage**.

Evaluation Metrics: This is commonly used in the detection tasks. For the OSSOD task, we adopt mean average precision (mAP), a standard evaluation metric widely used in object detection, to measure the performance of the proposed method on detecting ID samples, and Average Precision (AP_u) to evaluate its performance on OOD samples. To assess OOD detection, we use the area under the receiver operating characteristic curve (AUROC) and the FP rate at N% true positive rate (FPRN), which measure the method's ability to distinguish between ID and OOD samples.

Baseline: We adopt the previous classic SSOD approach [28] as the “baseline” model for comparison, to effectively validate the proposed method.

B. Main Results

1) *Performance on OSSOD:* Similar to [4], we first conduct experiments to evaluate the effectiveness of our proposed method on OSSOD tasks with varying numbers of ID categories and different levels of supervision, as shown in Table V. Compared to the existing methods, UT [37] and OF-DINO [4], our method shows significant improvements across different ID category counts and amounts of labeled data. Additionally, we conduct experiments on large-scale OSSOD benchmarks, namely COCO-OpenImage. DCH [63] is a recent method developed to address OSSOD tasks. Unlike our approach, which leverages OOD samples to enhance feature learning, DCH [63] follows OF-DINO [4] in filtering out anomalous samples during semi-supervised learning. The comparison results in Table VI show that our method consistently outperforms existing methods, including UT [37], OF-DINO [4], and DCH [63]. In Table VII, our approach performs in a manner that is comparable to the most advanced techniques [10] in ID class detection, while performing better in OOD detection, which is attributed to our effective MAM mechanism. The results further demonstrate that our approach significantly improves the performance of OSSOD algorithms in various scenarios, highlighting the effectiveness of appropriately utilizing OOD samples.

2) *Performance on OOD Detection:* To directly verify the efficacy of the prototype matching strategy we propose in

¹<https://github.com/open-mmlab/mmdetection>

²<https://pytorch.org/>

TABLE V

COMPARISON RESULTS OF OUR PROPOSED METHOD AND PREVIOUS WORKS ON COCO-OPEN USING MAP UNDER VARYING NUMBERS OF ID OBJECTS (4K LABELED IMAGES FROM 20, 40, AND 60 ID CLASSES) AND DIFFERENT LEVELS OF SUPERVISION (1K, 2K, AND 4K LABELED IMAGES FROM 20 SELECTED ID CLASSES). “**SUPERVISED**” REFERS TO TRAINING THE MODEL USING ONLY LABELED SAMPLES. BOTH **UT [37]** AND **OF-DINO [4]** ARE REPRODUCED IN OUR EXPERIMENTS UNDER THE SAME EXPERIMENTAL SETUPS AS OURS. EACH METHOD IS RUN THREE TIMES, AND WE REPORT THE MEAN AND STANDARD DEVIATION OF THE RESULTS

Method	ID/OOD Splits			Labeled Images		
	20/60	40/40	60/20	1K	2K	4K
Supervised	17.04±1.82	16.12±0.63	16.94±0.70	10.96±0.47	12.13±0.49	17.21±0.38
UT [37]	18.91±1.31	21.55±0.72	23.73±0.48	11.90±0.44	13.75±0.82	18.68±0.63
OF-DINO [4]	22.75±1.88	23.29±0.37	25.54±0.47	15.97±0.92	18.22±0.54	23.37±0.71
Ours	25.82±1.72	26.13±0.64	28.85±0.49	17.63±0.44	20.62±0.82	25.79±1.23

TABLE VI

COMPARISON RESULTS OF OUR PROPOSED METHOD AND PREVIOUS WORKS ON LARGE-SCALE BENCHMARK PROTOCOL COCO-OPENIMAGES USING MAP. “**SUPERVISED**” REFERS TO TRAINING THE MODEL USING ONLY LABELED SAMPLES. BOLD INDICATES THE BEST RESULTS

Method	Labeled	Unlabeled	mAP
Supervised	COCO	-	40.90
UT [37]	COCO	OpenImage	41.81
OF-DINO [4]	COCO	OpenImage	43.14
DCH [64]	COCO	OpenImage	44.13
Ours	COCO	OpenImage	44.19

TABLE VII

COMPARISON RESULTS OF OUR PROPOSED METHOD AND PREVIOUS WORKS ON VOC-COCO USING MAP AND AP_u AS METRICS. BOLD INDICATES THE BEST RESULTS, AND UNDERLINED INDICATES SUBOPTIMAL

Method	Labeled	Unlabeled	mAP	AP_u
UT [37]	VOC	COCO	28.82	0
DCH [64]	VOC	COCO	30.29	0
CFL [10]	VOC	COCO	31.49	<u>7.86</u>
Ours	VOC	COCO	<u>31.16</u>	8.53

this work for OOD sample detection, we present experimental results in Table VIII, comparing the proposed method with OE [64], OVA [46], Mahalanobis [68], Energy [65], GODIN [66], and OpenDet [67]. Additionally, we compare the multimetric matching mechanism with single-metric approaches, such as Euclidean distance and cosine similarity. In the proposed method, the matching score between an object and its prototype is treated as the probability that the object belongs to an ID sample. Notably, the prototypes updated through EMA exhibit strong class representativeness during training, and the multimetric matching mechanism provides a more comprehensive reflection of the similarity between features and prototypes. As shown in Table VIII, the proposed method outperforms other detection methods and single-metric matching mechanism, further validating the advantage of the proposed method in detecting OOD samples.

TABLE VIII

COMPARISON RESULTS OF OUR PROPOSED METHOD AND PREVIOUS WORKS FOR OOD DETECTION ON COCO-OPEN, WITH MODEL TRAINING USING 4K LABELED IMAGES FROM 20 SELECTED ID CLASSES. BOLD INDICATES THE BEST RESULTS

Method	Metric	AUROC	FPR75	FPR95
OE [65]	-	67.4	57.3	94.6
OVA [46]	-	72.7	46.5	89.2
Energy [66]	-	76.2	37.6	82.1
GODIN [67]	-	77.5	34.2	78.3
OpenDet [68]	-	69.9	54.1	87.7
OF-DINO [4]	-	83.7	19.6	55.4
Prototype-based Matching	Euclidean	79.3	31.8	70.1
	Cosine Similarity	81.1	25.7	58.6
	Mahalanobis [69]	74.6	38.4	83.5
	Ours	85.8	16.1	52.9

3) *Comparison With SSOD Methods:* Beyond the OSSOD task, we also evaluate the proposed approach against other state-of-the-art SSOD methods (i.e., STAC [23], UT [37], Soft Teacher [28], DCST [2], and PseCo [41]) under open-set conditions. As shown in Table IX, when SSOD algorithms are applied to open-set scenarios, existing techniques exhibit varying degrees of vulnerability to the adverse effects of OOD samples, leading to significant performance degradation. In contrast, our proposed method identifies OOD samples from unlabeled data and explicitly leverages them to facilitate feature learning and enhance the discrimination between ID and OOD samples, outperforming other SSOD methods in open-set scenes. Moreover, the proposed approach achieves results comparable to state-of-the-art methods on conventional SSOD tasks, which highlights the robustness of our method.

C. Ablation Studies

We conducted ablation studies on each individual component of the proposed method to assess their respective effectiveness. Unless specified otherwise, experiments were performed under the default setting of 40 known ID classes and 4K labeled images using the COCO-Open protocol.

1) *Instance-Level CR:* We first evaluate the contribution of ICR as an alternative for the traditional pseudo-labeling mechanism in the proposed method. By comparing results #2 with #1 and #7 with #5 in Table II, it can be observed

TABLE IX

COMPARISON RESULTS OF OUR PROPOSED METHOD AND OTHER SSOD METHODS USING MAP. **OPEN-SET** REFERS TO THE COCO-OPEN PROTOCOL WITH 40 ID AND 40 OOD CLASSES, USING 4K LABELED IMAGES. **CLOSED-SET (5%)** REFERS TO THE CASE WHERE 5% OF COCO (5850 IMAGES) IS USED AS LABELED DATA. **"SUPERVISED"** REFERS TO TRAINING THE MODEL USING ONLY LABELED SAMPLES. BOLD INDICATES THE BEST RESULTS

Method	Close-set (5%)	Open-set
Supervised	20.92	15.98
STAC [23]	24.38(+3.46)	18.97 (+2.99)
UT [37]	28.27(+7.35)	20.28(+4.30)
Soft Teacher [28]	30.74(+9.82)	20.98(+5.00)
DCST [2]	32.10 (+11.18)	22.97(+6.99)
PseCo [41]	32.50(+11.58)	23.01(+7.03)
Ours	32.38(+11.46)	26.13(+10.15)

that incorporating CR improves the model's detection performance, measured by mAP, by + 1.96 and + 1.41, respectively. This indicates that in OSSOD tasks, OOD samples with visual attributes similar to ID categories should be retained during training rather than discarded. Additionally, we use the weights w_j produced by MAM to regulate the consistency loss. As shown by comparing #4 with #2 in Table II, this effectively mitigates the negative impact of over-learned OOD samples on the learning process, further enhancing model performance. This study is the first to demonstrate the effectiveness of employing soft CR in the case of the tasks of object detection under an open-set semi-supervised scenario.

2) *OOD-Aware Contrastive Learning*: To verify the effectiveness of OCL, we perform comparison experiments where contrastive loss is used as the only incremental loss enabled. By comparing results #3 with #1 and #7 with #4 in Table II, we observe mAP gains of + 1.75 and + 1.68, respectively. This indicates that using $\mathcal{L}_{con}^l$ alone or both $\mathcal{L}_{con}^l$ and $\mathcal{L}_{con}^u$ contributes to improved performance over the baseline model.

To better understand the role of OCL, we visualize the learned features with and without the proposed OOD-aware contrastive loss, as shown in Fig. 4. With the introduction of the OOD-aware mechanism, OOD samples that are entangled with ID objects are pushed outside the class-wise ID feature clusters, significantly enhancing the discrimination between ID and OOD objects. This relieves the problem of overconfident predictions on OOD samples and improves the model's learning effects during training.

It is important to note that CR and contrastive learning are independent components, each capable of independently achieving notable performance improvements.

3) *Choice of Hyperparameters in Contrastive Learning*: Considering the "uniformity-tolerance dilemma" discussed in [72], we explored the effect of different temperature coefficients τ on the performance of the contrastive learning branch. As shown in Table X, when the temperature coefficient τ exceeds 0.2, the performance improvements from the contrastive loss remain stable. This stability may be due to the explicit hard negative mining, which ensures uniformity in our construction of sample pairs for contrastive learning.

TABLE X

ABLATION STUDY RESULTS OF THE PROPOSED METHOD WITH DIFFERENT CONTRASTIVE LEARNING BRANCH SETTINGS, EVALUATED USING MAP

D_C	Temperature Coefficient (τ)				
	0.05	0.1	0.2	0.4	0.7
128	24.84	25.58	26.11	26.02	26.08
256	24.90	25.49	26.07	25.98	26.01

TABLE XI

COMPARISON OF PARAMETER COUNTS BETWEEN THE PROPOSED MATCHING BRANCH AND THE PREVIOUS OF-DINO METHOD [4]

Method	Parameter Count
DINO (OF-DINO [4])	85M
Matching branch (Ours)	5.4M

Additionally, the results in Table X show that the proposed model's detection performance is not highly sensitive to the feature representation dimension. Thus, we set D_C to 128 to save computational resources.

4) *Multimetric Adaptive Matching*: To investigate how the MAM mechanism functions in the proposed algorithm, we analyze the model performance using mAP as the metric, specifically comparing #4 with #2 and #5 with #3 in Table II. The results suggest that the MAM mechanism has the potential to enhance both CR and contrastive learning, thereby maximizing their combined benefits for performance improvement. In addition, the comparison between #7 and #6 indicates that training the matching branch solely with labeled data results in a significant performance decline, underscoring the necessity of incorporating $\mathcal{L}_{mat}^u$. Additionally, it is important to highlight that the proposed matching branch consists of only a few lightweight fully connected layers, incurring negligible additional computational overhead compared to the backbone network, thus ensuring the efficiency of the overall end-to-end framework. This efficiency advantage is further validated in Table XI, which demonstrates that the parameter count of our matching branch is significantly lower than that of the existing OF-DINO approach [4], underscoring our method's computational effectiveness.

5) *Hyperparameter Sensitivity to D_M , σ_s and σ_l* : We also conduct analysis on hyperparameter sensitivity to validate the effects of the metric quantity D_M and the thresholds σ_l , σ_s , and σ_m . Regarding the metric quantity D_M , as shown in Fig. 5(a), the model performance in object detection (measured by mAP) and OOD detection (measured by AUROC) initially increases rapidly with the increase of D_M , then slowly declines after reaching a peak at $D_M = 5$. Additionally, concerning the impact of the thresholds σ_l , σ_s , and σ_m , empirical studies demonstrate that utilizing multiple metrics allows for a more comprehensive mapping of similarity between features, thereby surpassing mechanisms that rely on a single metric. The relatively poor performance observed in Fig. 5(a) indicates that the single threshold σ_m has inherent limitations in balancing the richness and reliability of contrastive samples.

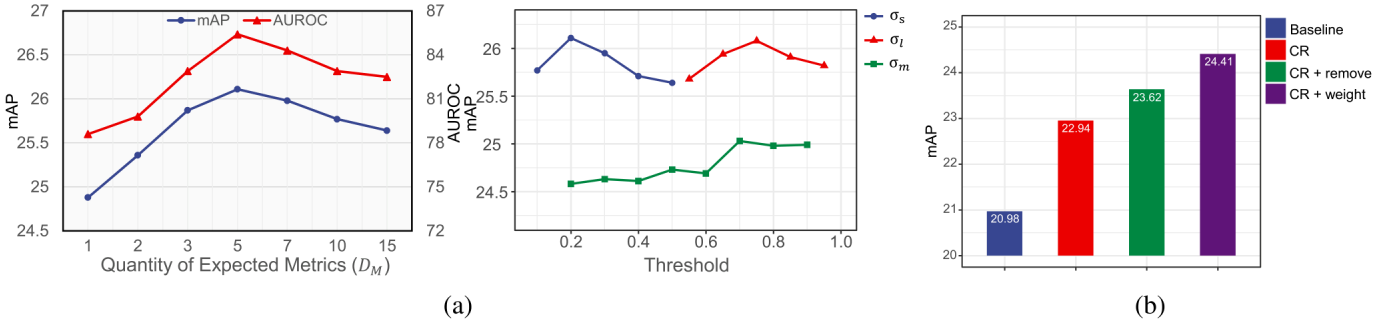


Fig. 5. (a) Hyperparameter sensitivity to D_M , σ_s and σ_l . (b) Performance comparisons of four different strategies for handling OOD samples in unlabeled data, conducted without the inclusion of the proposed contrastive learning approach.

Consequently, we adopt two thresholds, σ_l and σ_s , to filter ID and OOD samples separately. By imposing these two thresholds, ambiguous samples with matching scores between σ_l and σ_s are disregarded due to their uncertainty, ensuring the reliability of contrastive samples and consequently enhancing overall performance. Optimal performance is achieved at $\sigma_l = 0.75$ and $\sigma_s = 0.2$.

6) *Removing OOD Samples Versus Weighting Control:* Effectively handling OOD samples is crucial to the success of the OSSOD task. We explore four different strategies for processing OOD samples in unlabeled data, with the results presented in Fig. 5(b). These strategies, from left to right, are: pseudo-labeling in the baseline model, using ICR instead of pseudo-labeling, setting CR weights to zero for OOD objects (with matching scores lower than σ_s) identified by MAM, and replacing CR weights with matching scores w_j for OOD objects identified by MAM.

The comparison between Bar 2 and Bar 1 demonstrates the superior robustness of CR in open-set scenarios compared to pseudo-labeling. When comparing Bar 3 and Bar 2, we observe that removing OOD samples identified by MAM leads to a further performance improvement, confirming the negative impact of OOD samples on learning ID categories. Additionally, as shown in the comparison between Bar 4 and Bar 3, the mAP increases by + 0.79, indicating that the semantic information within OOD samples retains value for feature learning. This supports our hypothesis that appropriately leveraging OOD samples can enhance feature representation and improve detection performance.

D. Qualitative Results

1) *Visualization of Instance-Level CR:* Pseudo-labeling is inherently unsuitable for SSOD in open-set scenes. Lowering the threshold for pseudo-labeling increases the presence of inaccurate pseudo-labels, which can lead to multiple noisy training instances through IoU matching. Conversely, raising the threshold limits the recall of training instances, resulting in many ID foreground objects being misclassified as background. As a result, pseudo-labeling in SSOD for open-set scenarios heavily depends on advanced OOD detectors.

To address this, we propose an enhanced CR, as shown in Fig. 6, which reduces the reliance on high-confidence pseudo-labels and increases the availability of foreground object boxes during training. Additionally, CR demonstrates robust handling

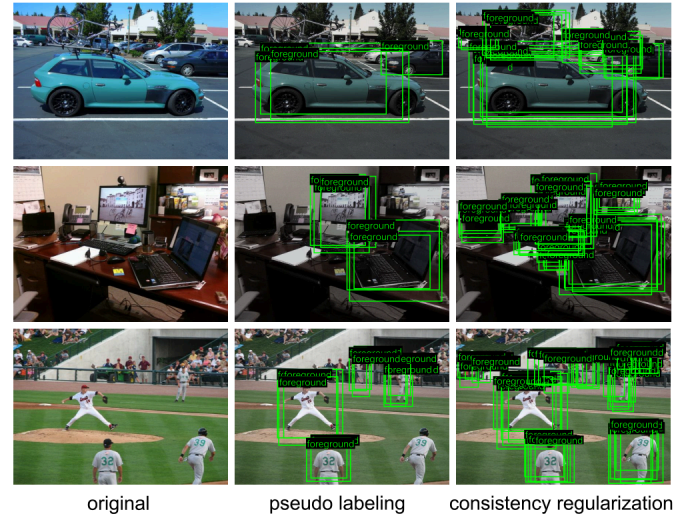


Fig. 6. Visualization of foreground boxes for pseudo-labeling and the proposed consistency learning during the training process.

of OOD samples, significantly improving the performance of the detector.

2) *Visualization of OOD Detection:* To demonstrate the effectiveness of our proposed method in OOD detection, we visualized the qualitative results of both the original and corrected outputs. As illustrated in Fig. 7, in the original output, some OOD objects (e.g., butterfly) were incorrectly predicted as ID objects (e.g., bird) with high confidence. By optimizing the feature space through OCL, the OOD features in unlabeled data become more distinguishable via a MAM mechanism. These features are subsequently utilized for representation learning. The interaction among multiple modules reflects our effective exploration and utilization of OOD samples, thereby enhancing the accuracy of the object detector. Notably, the bottom row of Fig. 7 presents representative failure cases of our proposed method. Similar to other classic methods [4], [28], [37], this proposed approach struggles when OOD objects share similar background knowledge with ID objects (e.g., as shown in Fig. 7, the habitat of the OOD-labeled “sheep” is very similar to the ID-labeled “horse”). Additionally, our proposed method is prone to failure when the image contains a complex number of objects to detect. This highlights a challenge we aim to thoroughly address in future work.

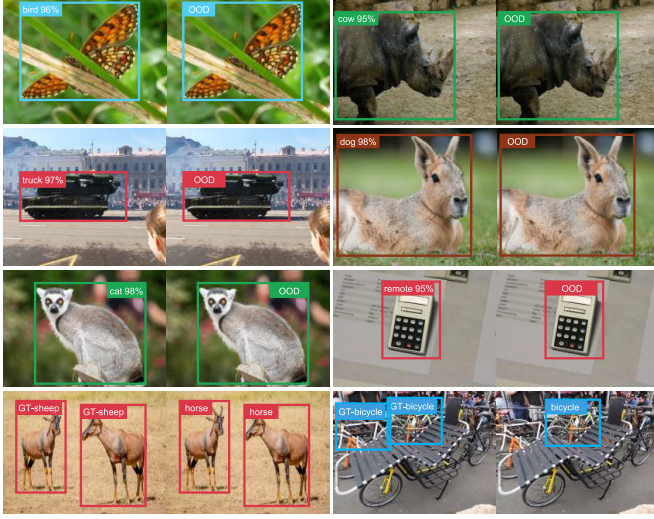


Fig. 7. Qualitative results illustrating the effectiveness of our proposed method in distinguishing OOD samples from ID categories. In the first three rows, the left side shows the incorrect results from the original Faster R-CNN, where OOD objects are wrongly predicted as ID classes with high confidence. The right side presents the corrected results after applying our MAM method. The bottom row displays some representative failure cases. On the bottom-left, the ground-truth label of a sheep (OOD class) is incorrectly predicted as a horse (ID class) by our model. On the bottom-right, a densely packed scene is shown, where our model missed a bicycle object.

V. THEORETICAL ANALYSIS AND CONVERGENCE GUARANTEES

In this section, we present rigorous theoretical justifications for our proposed method for the task of OSSOD, specifically focusing on ICR, OCL, and MAM. The joint optimization objective used in our approach is

$$\mathcal{L}(\theta) = \mathcal{L}_{det} + \gamma \mathcal{L}_{con} + \mathcal{L}_{mat}$$

where symbols are consistent with the definitions as stated above.

A. Theoretical Assumptions

We adopt the following assumptions consistent with the experimental conditions described in the main manuscript.

- A1 *Gaussian Distribution of Features*: The feature embeddings from ID and OOD objects form Gaussian clusters with bounded variance and separation margin Δ .
- A2 *Lipschitz Continuity of Augmentations*: Weak and strong augmentations $\mathcal{A}^W, \mathcal{A}^S$ are Lipschitz continuous transformations with Lipschitz constant L .
- A3 *Prototype Boundedness*: Class prototypes updated via EMA are bounded by a radius R .
- A4 *Learning Rate Constraint*: The SGD learning rate η satisfies $\eta < 1/\lambda_{\max}(H)$, where H is the Hessian matrix of $\mathcal{L}_{det}$.

These assumptions align well with our empirical observations and practical implementations.

B. Key Lemmas

We introduce important lemmas that underpin our main results. Complete proofs are in Section VI.

Lemma 1 (Smoothness and Stability of CR): Under assumptions A2–A4, the consistency loss $\mathcal{L}_{det}$ is $2L$ -smooth, and SGD optimization with step size $\eta < 1/(2L)$ is uniformly stable with stability constant $O(L^2/n)$.

Lemma 2 (Margin Expansion by OCL): Each gradient step optimizing $\mathcal{L}_{con}$ increases the ID/OOD margin γ_t by at least

$$\Delta\gamma_t \geq \frac{\eta\gamma}{\tau} (\gamma_t - \gamma_{\min}), \quad \gamma_{\min} = 1 - \sigma/\Delta$$

where σ denotes a parameter that controls the separation or margin between the ID and OOD classes in the feature space.

Lemma 3 (Gradient Robustness from MAM): The MAM mechanism with weight $w_j < \sigma_s$ scales the gradient contribution of OOD samples, preventing significant gradient drift.

These lemmas collectively assure robust training and explicit margin expansion.

C. Main Theoretical Results

Our primary theoretical results demonstrate stable convergence and improved generalization.

Theorem 1 (Global Convergence Guarantee): Under assumptions A1–A4, alternating SGD optimization of (V) converges to a solution θ^* , with margin $\gamma^* \geq \gamma_{\min}$.

Theorem 2 (Generalization Risk Bound): With probability at least $1 - \delta$, the generalization risk $\mathcal{R}(\theta^*)$ satisfies

$$\mathcal{R}(\theta^*) \leq O\left(\frac{L^2}{n}\right) + O\left(e^{-\gamma^*}\right) + O\left(\sqrt{\frac{\log(1/\delta)}{n+m}}\right)$$

where n, m are the counts of labeled and mined unlabeled instances, respectively. Here, δ represents the failure probability of the model's generalization risk, meaning the model will achieve the given generalization error bound with a probability of $(1 - \delta)$.

These theorems confirm theoretical validity and explain empirical performance gains.

VI. FORMAL PROOFS, EMPIRICAL VALIDATION, AND DISCUSSION

In this section, we provide detailed proofs for theoretical results in Section V-C, empirical validation experiments, and discuss the limitations and future directions of our method.

A. Proof of Lemma 1

The CR loss $\mathcal{L}_{det}$ is a smooth composition of Lipschitz continuous augmentations and stable cross-entropy losses. Therefore, following standard smoothness and stability results [73], we directly obtain the stated smoothness and stability bound. $\square$

B. Proof of Lemma 2

Considering the InfoNCE-based formulation of our contrastive loss, we compute gradients explicitly

$$\nabla_z \mathcal{L}_{con} \propto -\frac{\mu_c - \cos(z, \mu_c)z}{\tau}.$$

A single-step gradient update results in a margin increase bounded as

$$\gamma_{t+1} - \gamma_t \geq \frac{\eta\gamma}{\tau} (\gamma_t - \gamma_{\min})$$

confirming explicit margin enhancement. $\square$

TABLE XII
MARGIN (γ^*) VERSUS DETECTION PERFORMANCE (mAP)

Margin γ^*	0.1	0.2	0.3	0.5
mAP	22.73	24.68	25.82	26.13

C. Proof of Lemma 3

For OOD instances identified by MAM, the weighted gradient is scaled down explicitly by w_j , thus effectively preventing substantial interference from OOD samples

$$\|\nabla(w_j \ell)\| \leq w_j \|\nabla \ell\| \leq \sigma_s \|\nabla \ell\|, \quad w_j < \sigma_s.$$

□

D. Proof of Theorem 1

We define a Lyapunov function incorporating Lemmas 1–3 and employ standard SGD convergence theory [73], which yields global convergence to a stationary point θ^* with guaranteed margin. □

E. Proof of Theorem 2

Combining Lemmas 1, 2, and uniform stability results with standard margin-based generalization error theory [74], we achieve the stated risk bound. □

F. Empirical Validation of Margin Influence

Table XII empirically confirms the theoretical relationship: explicitly improving the margin γ^* (controlled by temperature parameter τ) reduces the out-of-sample detection error significantly.

G. Limitations and Future Directions

Our theoretical analysis currently relies on simplified assumptions such as single-mode Gaussian distributions and linear approximations. Extending theoretical guarantees to multimodal OOD distributions, non-Gaussian embedding clusters, and deriving tighter nonconvex generalization bounds represent meaningful directions for future work. Empirical exploration of adaptive hyperparameter tuning and further scalability analysis would also strengthen practical applicability.

VII. LIMITATIONS AND FUTURE WORK

Our method delivers strong performance on OSSOD, and to guide deployment, we outline its practical scope in everyday terms. Separating very similar ID and OOD categories can still be demanding, especially in fine-grained cases or scenes with occlusion, cluttered backgrounds, or only partial views; in such conditions, the margin over simpler detectors can narrow, particularly in time-critical settings such as autonomous driving and crowded-scene surveillance. Performance can also taper when the labeled and unlabeled data differ greatly—for example, moving from daytime imagery to nighttime footage, or from visible-light video to infrared video—because feature statistics and appearance cues change substantially. In

addition, training benefits from accurate and consistent labels; significant annotation noise can reduce the effectiveness of supervision and slow convergence. We view these not as drawbacks but as concrete opportunities for improvement: incorporating stronger instance-level consistency to stabilize learning under occlusion and partial visibility, adding targeted domain-adaptation modules to bridge large appearance shifts, and introducing lightweight data-quality checks or noise-robust training objectives to maintain reliability when labels are imperfect. These steps help clarify the operating envelope of the method and support more confident, predictable deployment in real-world applications.

VIII. CONCLUSION

In this article, we contribute instance-level CR and OCL to tackle the task of OSSOD. In particular, the instance-level CR robustly enhances the recognition capability by facilitating the learning of object feature invariance across various disturbances, while the OCL optimizes the feature learning of the model by leveraging OOD features to aggravate the distinction between ID and OOD objects. Furthermore, we present a novel prototype-based MAM to identify ID/OOD objects in unlabeled data, which improves the learning approach. The results of numerous experiments demonstrate that the proposed approach consistently surpasses existing methods, which indicates that using OOD samples to ameliorate feature representation is crucial.

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